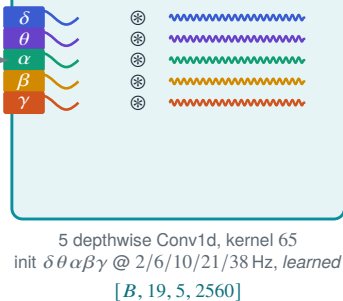


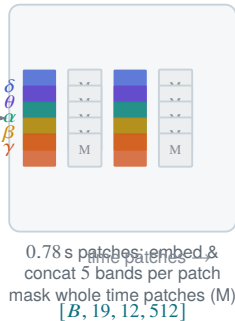
each band = input $\otimes$ its
learnable kernel h_b

Learnable filterbank

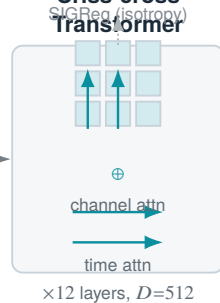
Raw EEG
[B, 2560, 19] 10 s $\times$ 19 ch



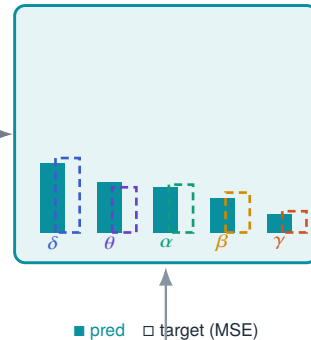
Band tokens + mask time



Criss-cross Transformer



Predict band power



$\mathcal{L} = \mathcal{L}_{\text{sp}} + 0.05 \mathcal{L}_{\text{sig}}$
5-D band-power summary of the
masked patch — not its full
spectrum: lossy, denoised.
Head discarded after pretrain-
ing;
the encoder transfers.
no codebook · no reconstruc-
tion · no JEPA

Fixed filterbank h^{fix} (frozen)

$$\bar{e}_{c,t,b} = \frac{1}{P} \sum_{\tau} (h_b^{\text{fix}} * x)^2$$

target = $\log(1 + \bar{e})$

raw signal, no gradient

MSE target